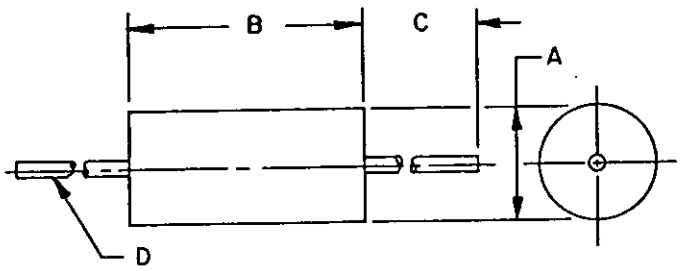
MILITARY SPECIFICATION SHEET

COILS, FIXED, RADIOFREQUENCY, MOLDED, MICRO-MINIATURE, MAGNETICALLY SHIELDED (PHENOLIC CORE-IRON SLEEVE), ESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-39010.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.152 (3.86)	.172 (4.37)
B	.390 (9.91)	.430(10.92)
C	1.250(31.75)	1.626(41.30)
D	.023 (.58)	.027 (.69)

NOTE: Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Material: Phenolic core - iron sleeve.

Weight: 0.850 gram maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202 test voltage: 1,000 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202, (70,000 feet), test voltage 200 volts rms.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Inductance tolerance: See table I.

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Q values: See table I.

Self-resonant frequency: See table I.

Percent coupling: 3 percent maximum.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part number: M39010/01 (dash number from table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
AR10**	.10	5,10	50	25	250	.025	1790
AR11**	.11	5	50	25	250	.034	1530
AR12**	.12	5,10	50	25	250	.034	1530
AR13**	.13	5	50	25	250	.037	1470
AR15**	.15	5,10	50	25	250	.037	1470
AR16**	.16	5	50	25	250	.047	1300
AR18**	.18	5,10	50	25	250	.047	1300
AR20**	.20	5	49	25	250	.067	1100
AR22**	.22	5,10	49	25	250	.067	1100
AR24**	.24	5	47	25	250	.11	853
AR27**	.27	5,10	47	25	250	.11	853
AR30**	.30	5	46	25	250	.13	780
AR33**	.33	5,10	46	25	250	.13	780
AR36**	.36	5	44	25	250	.18	670
AR39**	.39	5,10	44	25	250	.18	670
AR43**	.43	5	44	25	235	.25	565
AR47**	.47	5,10	44	25	235	.25	565
AR51**	.51	5	43	25	210	.33	490
AR56**	.56	5,10	43	25	210	.33	490
AR62**	.62	5	42	25	190	.45	420

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - *Continued.*

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
AR68**	.68	5,10	42	25	190	.45	420
AR75**	.75	5	40	25	180	.59	370
AR82**	.82	5,10	40	25	180	.59	370

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the failure rate symbol (e.g., M, P, R, S) and will be added to the end of the dash number.
2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from initial measurement		Allowable % from specified minimum value in electrical characteristics (initial) table	
	Inductance	DC resistance	Self-resonant frequency	Q
	Percent			
Qualification inspection				
Group II	±5	---	---	-10
Group IV	±5	±(3%+.001 ohm)	1/	-10
Group V	±5	±(2%+.001 ohm)	1/	-10
Quality conformance inspection				
Group C				
Subgroup 1	±5	±(2%+.001 ohm)	1/	-10
Subgroup 3	±5	---	---	-10
Subgroup 4	±5	±(3%+.001 ohm)	1/	-10

1/ The self-resonant frequency shall be not less than the value specified in table I.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, SH
Air Force - 19

Agent:
DLA-ES

Preparing activity:
Army - ER

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